LDA and Decision tree

Lecture 04

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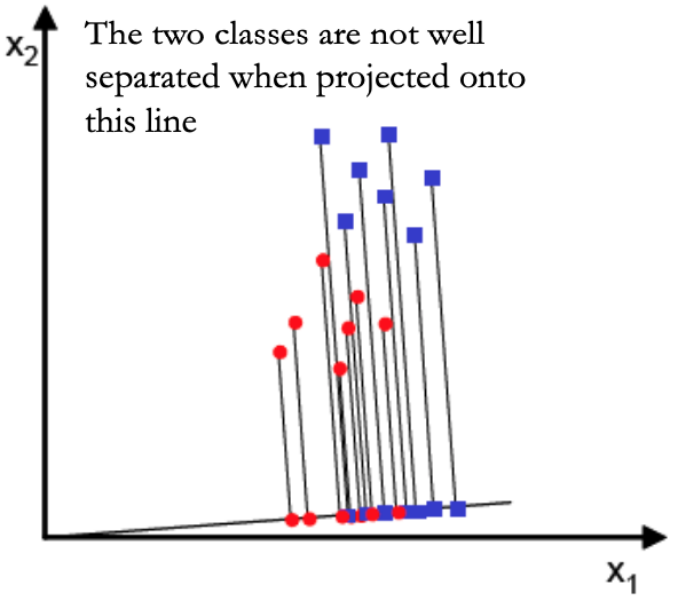
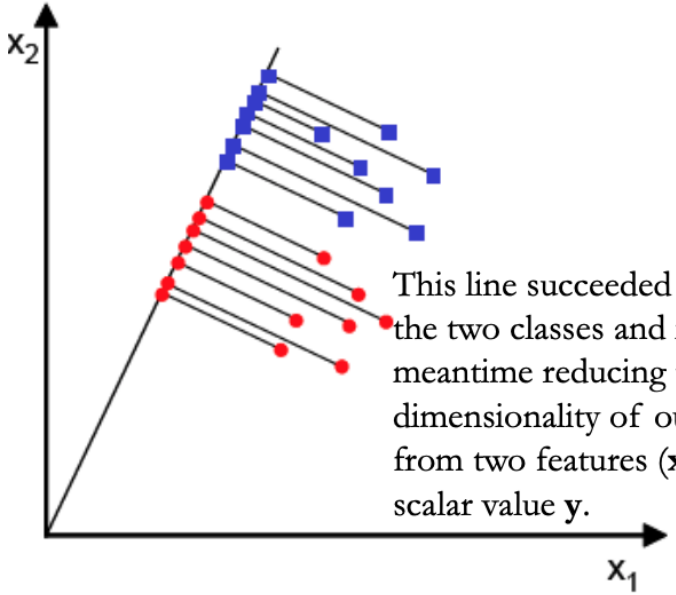
Outline

- **Linear Discriminant Analysis (LDA)**
- Decision tree (决策树)

Maximize the separation





This line succeeded in separating the two classes and in the meantime reducing the dimensionality of our problem from two features ($\mathbf{x}_1$,$\mathbf{x}_2$) to only a scalar value **y**.

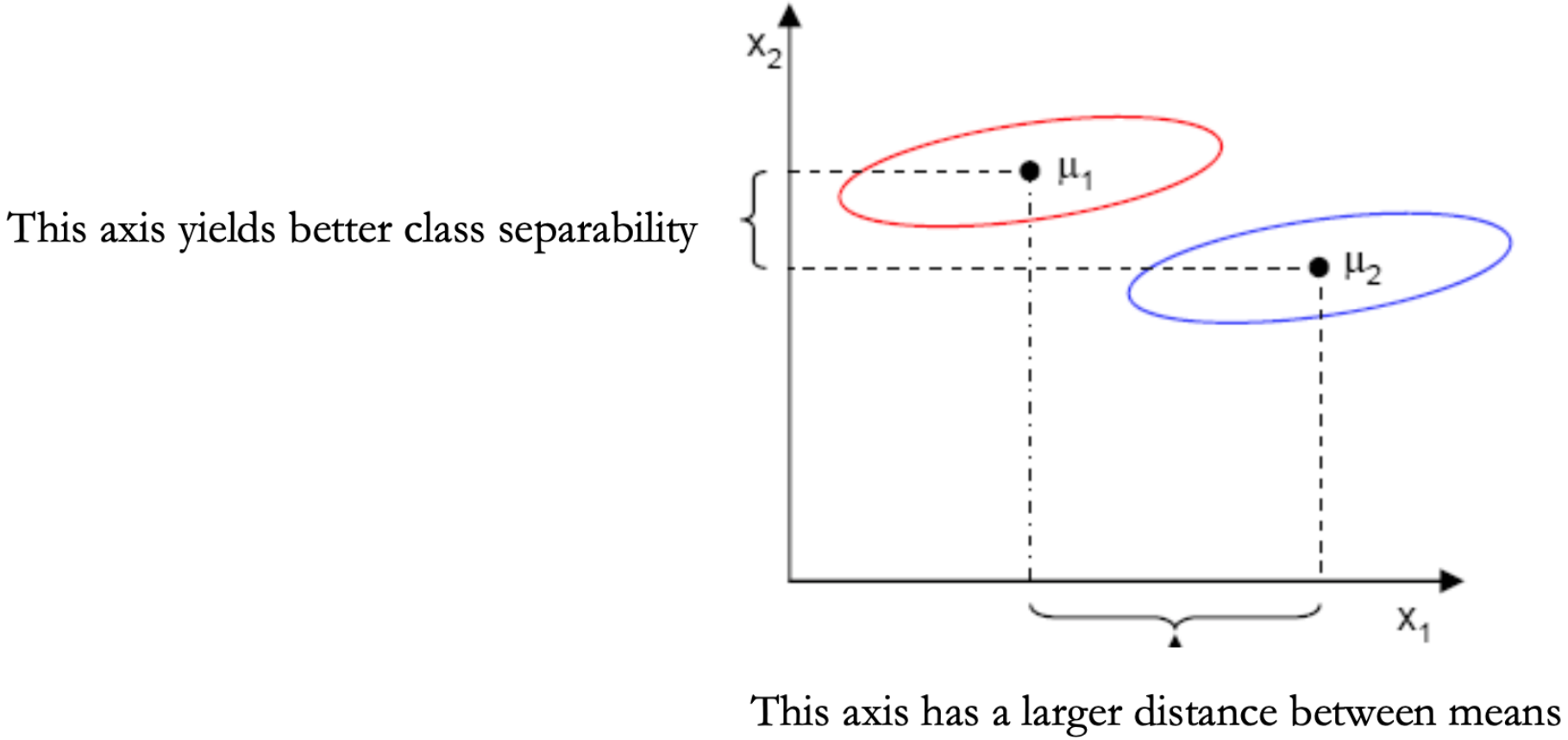
- Assume we have *m*-dimensional samples {$\mathbf{x}^1$, $\mathbf{x}^2$,…, $\mathbf{x}^N$}, N_1 of which belong to $\boldsymbol{\omega}_1$ and N_2 belong to $\boldsymbol{\omega}_2$.
- We seek to obtain a scalar **y** by projecting the samples **x** onto a line (C-1 space, C = 2).

$$y = w^T x \quad where \quad x = \begin{bmatrix} x_1 \\ . \\ . \\ x_m \end{bmatrix} \quad and \quad w = \begin{bmatrix} w_1 \\ . \\ . \\ w_m \end{bmatrix}$$

  - where **w** is the projection vectors used to project **x** to **y**.

- **Of all the possible lines we would like to select the one that maximizes the separability of the scalars.**

How to measure the separation?

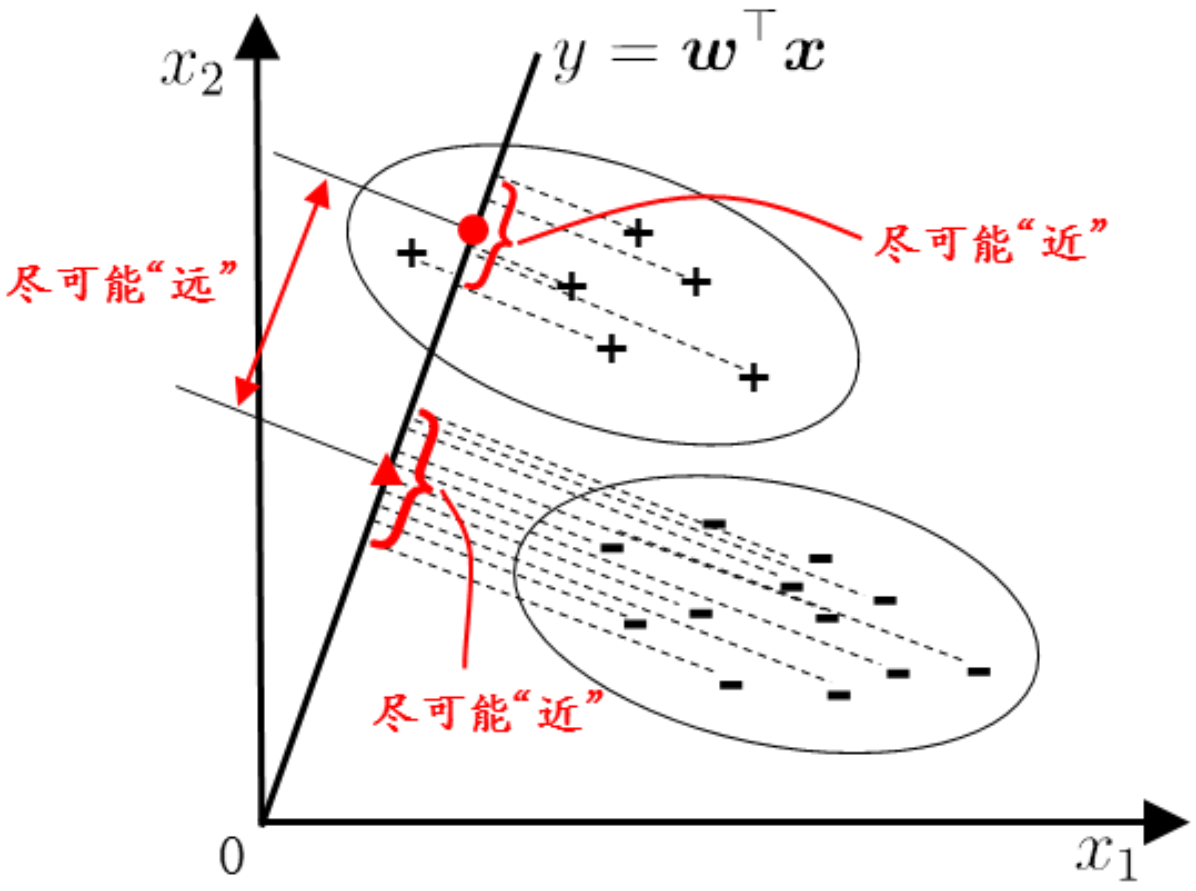


- The distance between the projected means is not a very good measure since it does not take into account the standard deviation within the classes.

The idea of linear discriminant analysis (LDA)

- 欲使**同类**样例的投影点尽可能**接近**，可以让同类样例投影点的协方差尽可能小

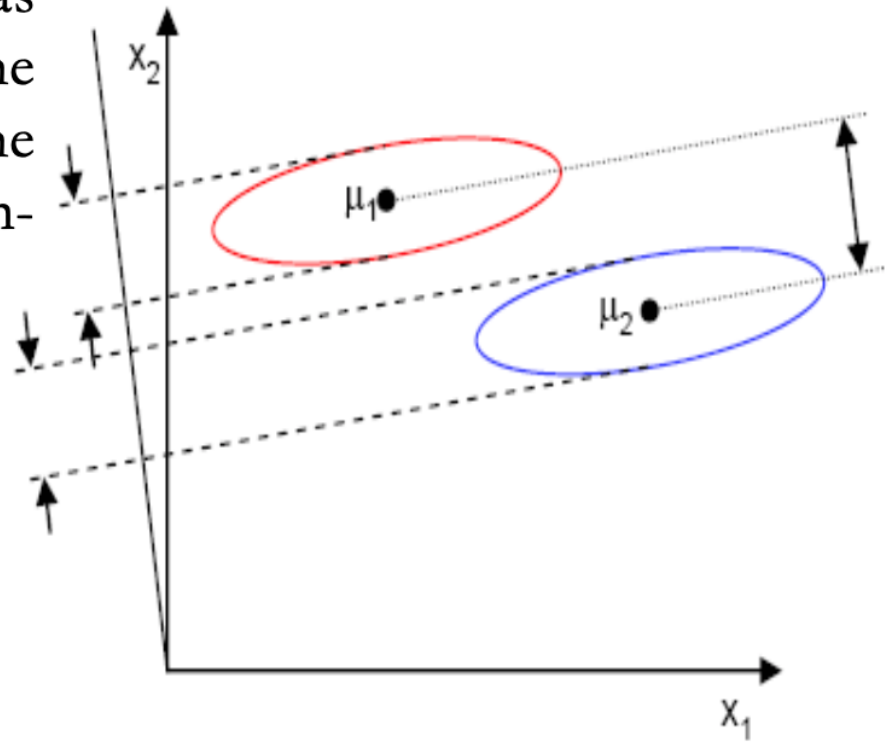
- 欲使**异类**样例的投影点尽可能**远离**，可以让不同类中心之间的距离尽可能大



Fisher's LDA criterion

- The Fisher linear discriminant is defined as the linear function $\mathbf{w}^T\mathbf{x}$ that maximizes the criterion function: (the distance between the projected means normalized by the within-class scatter of the projected samples.

$$J(w) = \frac{\left|\tilde{\mu}_1 - \tilde{\mu}_2\right|^2}{\tilde{s}_1^2 + \tilde{s}_2^2}$$



- Therefore, we will be looking for a projection where examples from the same class are projected very close to each other and, at the same time, the projected means are as farther apart as possible

Within-class scatter (类内散度)

- Now, the scatter of the projection **y** can then be expressed as a function of the scatter matrix in feature space **x**.

$$J(w) = \frac{\left|\tilde{\mu}_1 - \tilde{\mu}_2\right|^2}{\tilde{s}_1^2 + \tilde{s}_2^2}$$

$$\tilde{s}_i^2 = \sum_{y \in \omega_i} \left(y - \tilde{\mu}_i\right)^2 = \sum_{x \in \omega_i} \left(w^T x - w^T \mu_i\right)^2$$

$$= \sum_{x \in \omega_i} w^T \left(x - \mu_i\right)\left(x - \mu_i\right)^T w$$

$$= w^T \left(\sum_{x \in \omega_i} \left(x - \mu_i\right)\left(x - \mu_i\right)^T \right) w = w^T S_i w$$

$$\tilde{s}_1^2 + \tilde{s}_2^2 = w^T S_1 w + w^T S_2 w = w^T \left(S_1 + S_2\right) w = w^T S_W w = \tilde{S}_W$$

Where $\tilde{S}_W$ is the <u>within-class scatter matrix</u> of the projected samples **y**.

Between-class scatter （类间散度）

- Similarly, the difference between the projected means (in y-space) can be expressed in terms of the means in the original feature space (x-space).

$$\left(\tilde{\mu}_1 - \tilde{\mu}_2\right)^2 = \left(w^T\mu_1 - w^T\mu_2\right)^2$$

$$= w^T \underbrace{\left(\mu_1 - \mu_2\right)\left(\mu_1 - \mu_2\right)^T}_{S_B} w$$

$$= w^T S_B w = \tilde{S}_B$$

$$\boxed{J(w) = \frac{\left|\tilde{\mu}_1 - \tilde{\mu}_2\right|^2}{\tilde{s}_1^2 + \tilde{s}_2^2}}$$

- The matrix $\mathbf{S_B}$ is called the *between-class scatter* of the original samples/feature vectors, while $\tilde{S}_B$ is the between-class scatter of the projected samples $\mathbf{y}$.

- Since $\mathbf{S_B}$ is the outer product of two vectors, its rank is at most one.

The LDA's separation criterion

- We can express the Fisher criterion as:

$$J(w) = \frac{\left|\tilde{\mu}_1 - \tilde{\mu}_2\right|^2}{\tilde{s}_1^2 + \tilde{s}_2^2} = \frac{w^T S_B w}{w^T S_W w}$$

Hence J(w) is a measure of the difference between class means (encoded in the between-class scatter matrix) normalized by a measure of the within-class scatter matrix.

Solve LDA

$$\max_{\mathbf{w}} J(\mathbf{w}) = \frac{\mathbf{w}^T S_B \mathbf{w}}{\mathbf{w}^T S_W \mathbf{w}}$$

Since *J(w)* is invariant with respect to rescale of *w*, we can transform the problem of maximizing *J(w)* into the following constrained optimization problem,

$$\begin{aligned} \min_{\mathbf{w}} \quad & -\frac{1}{2}\mathbf{w}^T S_B \mathbf{w} \\ \text{s.t.} \quad & \mathbf{w}^T S_W \mathbf{w} = 1 \end{aligned}$$

Compute the Lagrangian

$$\mathcal{L}_P = -\frac{1}{2}\mathbf{w}^T S_B \mathbf{w} + \frac{1}{2}\lambda(\mathbf{w}^T S_W \mathbf{w} - 1)$$

From the KKT conditions, we have $S_B \mathbf{w} = \lambda S_W \mathbf{w} \quad \Rightarrow \quad S_W^{-1} S_B \mathbf{w} = \lambda \mathbf{w}$

Define $\mathbf{v} = S_B^{\frac{1}{2}} \mathbf{w}$, then $S_B^{\frac{1}{2}} S_W^{-1} S_B^{\frac{1}{2}} \mathbf{v} = \lambda \mathbf{v}$

It is an Eigenvalue problem!

From Two-Class to Multiple-Class

- Decompose a n-class problem into several two-classes problems:

- **One-vs-Rest**: Use n-1 binary classifiers each of which solves a two-class problem of separating points in a particular class from points not in that class.
  - **Pros**: small storing cost and short test time
  - **Cons**: long training time, imbalanced Samples.

- **One-vs-One**: Use $(n - 1)n/2$ binary classifiers, one for every possible pairs of classes. Each point is classified according to a majority vote amongst the discriminant functions.
  - **Pros**: short training time
  - **Cons**: Too many classifiers; large storing cost and long test time

Multi-class LDA

- Fisher LDA considers maximizing the following objective:

$$J(\mathbf{w}) = \frac{\mathbf{w}^T S_B \mathbf{w}}{\mathbf{w}^T S_W \mathbf{w}}$$

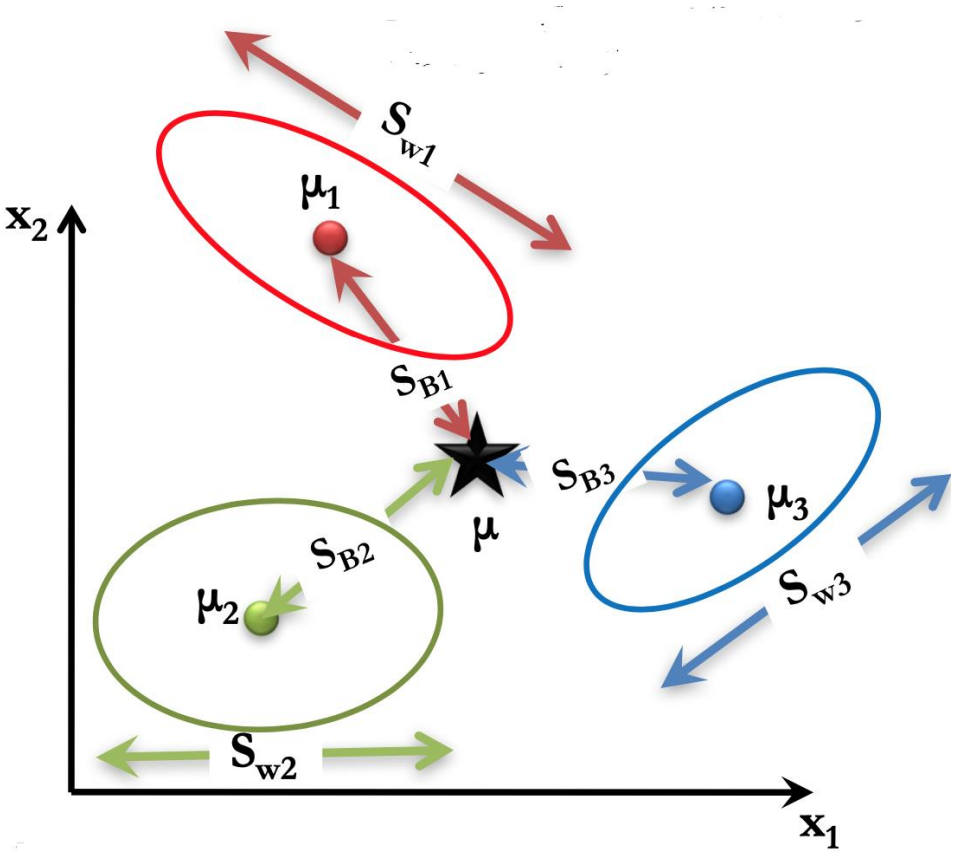
where

Between-class scatter matrix:

$$S_B = \sum_c (\boldsymbol{\mu}_c - \bar{\mathbf{x}})(\boldsymbol{\mu}_c - \bar{\mathbf{x}})^T$$

Within-class scatter matrix

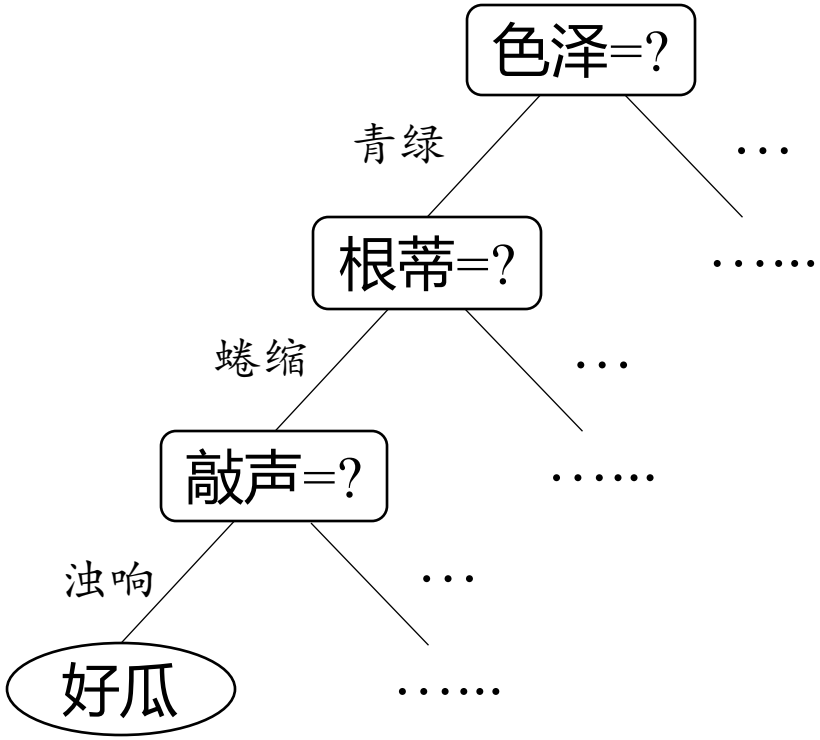
$$S_W = \sum_c \sum_{i \in c} (\mathbf{x}_i - \boldsymbol{\mu}_c)(\mathbf{x}_i - \boldsymbol{\mu}_c)^T$$



Outline

- Linear Discriminant Analysis (LDA)
- **Decision tree (决策树)**

决策树基于树结构来进行预测



基本流程

- 决策过程中提出的每个判定问题都是对某个属性的“测试”
- 决策过程的最终结论对应了我们所希望的判定结果
- 每个测试的结果或是导出最终结论，或者导出进一步的判定问题，其考虑范围是在上次决策结果的限定范围之内
- 从根结点到每个叶结点的路径对应了一个判定测试序列

决策树学习的目的是为了产生一棵泛化能力强，即处理未见示例能力强的决策树

基本流程

Algorithm 1 决策树学习基本算法

输入:

- 训练集 $D = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_m, y_m)\}$;
- 属性集 $A = \{a_1, \ldots, a_d\}$.

过程: 函数 TreeGenerate(D, A)

```
生成结点 node;
if D 中样本全属于同一类别 C then
    将 node 标记为 C 类叶结点; return
end if
if A = ∅ OR D 中样本在 A 上取值相同 then
    将 node 标记叶结点，其类别标记为 D 中样本数最多的类; return
end if
从 A 中选择最优划分属性 a*;
for a* 的每一个值 a*^v do
    为 node 生成每一个分枝; 令 D_v 表示 D 中在 a* 上取值为 a*^v 的样本子集;
    if D_v 为空 then
        将分枝结点标记为叶结点, 其类别标记为 D 中样本最多的类; return
    else
        以 TreeGenerate(D_v, A − {a*}) 为分枝结点
    end if
end for
```

输出: 以 node 为根结点的一棵决策树

（1）当前结点包含的样本全部属于同一类别

（2）当前属性集为空，或所有样本在所有属性上取值相同

（3）当前结点包含的样本集合为空

决策树算法的核心是要解决两个问题：

1）如何从数据集中找出最佳分枝?

2）如何在合适的时候让决策树停止生长，防止过拟合?

划分选择

- 决策树学习的关键在于如何选择最优划分属性。一般而言，随着划分过程不断进行，我们希望决策树的分支结点所包含的样本尽可能属于同一类别，即结点的“纯度”(purity)越来越高

- 经典的属性划分方法：
  - 信息增益
  - 增益率
  - 基尼指数

划分选择-信息增益

- “信息熵”是度量样本集合纯度最常用的一种指标，假定当前样本集合 D 中第 k 类样本所占的比例为 $p_k(K = 1,2,\dots,|Y|)$，则 D 的信息熵定义为

$$H(D) = \mathrm{Ent}(D) = -\sum_{k=1}^{|Y|} p_k \log p_k$$

  $H(D)$ 的值越小，则 D 的纯度越高。
- 计算信息熵时约定：若 $p = 0$，则 $p\log(p) = 0$
- $H(D)$ 的最小值为 0，最大值为 $\log|Y|$

划分选择-信息增益

- 离散属性 a 有 V 个可能的取值 $\{a^{(1)}, a^{(2)}, \ldots, a^{(V)}\}$，用 a 来进行划分，则会产生 V 个分支结点，其中第 v 个分支结点包含了 D 中所有在属性 a 上取值为 $\mathrm{a}^{(v)}$ 的样本，记为 D^v。则可计算出用属性 a 对样本集 D 进行划分所获得的“信息增益”：

$$\mathrm{Gain}(D, a) = \mathrm{Ent}(D) - \sum_{v=1}^{V} \frac{|D^v|}{|D|} \mathrm{Ent}(D^v)$$

为分支结点权重，样本数越多的分支结点的影响越大

- 一般而言，**信息增益越大**，则意味着使用属性 来进行划分所获得的“**纯度提升**”越大
- ID3决策树学习算法[Quinlan, 1986]以信息增益为准则来选择划分属性

信息增益实例

编号	色泽	根蒂	敲声	纹理	脐部	触感	好瓜
1	青绿	蜷缩	浊响	清晰	凹陷	硬滑	是
2	乌黑	蜷缩	沉闷	清晰	凹陷	硬滑	是
3	乌黑	蜷缩	浊响	清晰	凹陷	硬滑	是
4	青绿	蜷缩	沉闷	清晰	凹陷	硬滑	是
5	浅白	蜷缩	浊响	清晰	凹陷	硬滑	是
6	青绿	稍蜷	浊响	清晰	稍凹	软粘	是
7	乌黑	稍蜷	浊响	稍糊	稍凹	软粘	是
8	乌黑	稍蜷	浊响	清晰	稍凹	硬滑	是
9	乌黑	稍蜷	沉闷	稍糊	稍凹	硬滑	否
10	青绿	硬挺	清脆	清晰	平坦	软粘	否
11	浅白	硬挺	清脆	模糊	平坦	硬滑	否
12	浅白	蜷缩	浊响	模糊	平坦	软粘	否
13	青绿	稍蜷	浊响	稍糊	凹陷	硬滑	否
14	浅白	稍蜷	沉闷	稍糊	凹陷	硬滑	否
15	乌黑	稍蜷	浊响	清晰	稍凹	软粘	否
16	浅白	蜷缩	浊响	模糊	平坦	硬滑	否
17	青绿	蜷缩	沉闷	稍糊	稍凹	硬滑	否

该数据集包含17个训练样本，$|\mathcal{Y}| = 2$，其中正例占 $p_1 = \frac{8}{17}$，反例占 $p_2 = \frac{9}{17}$，计算得到根结点的信息熵为

$$\mathrm{Ent}(D) = -\sum_{k=1}^{2} p_k \log_2 p_k = -(\frac{8}{17}\log_2\frac{8}{17} + \frac{9}{17}\log_2\frac{9}{17}) = 0.998$$

信息增益实例

- 以属性“色泽”为例，其对应的3个数据子集分别为 D^1(色泽=青绿)，D^2(色泽=乌黑)，D^3 (色泽=浅白)
- 子集 D^1 包含编号为$\{1,4,6,10,13,17\}$的6个样例，其中正例占 $p_1=\frac{3}{6}$ ，反例占 $p_2=\frac{3}{6}$ ，D^2、D^3 同理，3个结点的信息熵为：

$$\mathrm{Ent}(D^1)=-(\tfrac{3}{6}\log_2\tfrac{3}{6}+\tfrac{3}{6}\log_2\tfrac{3}{6})=1.000$$

$$\mathrm{Ent}(D^2)=-(\tfrac{4}{6}\log_2\tfrac{4}{6}+\tfrac{2}{6}\log_2\tfrac{2}{6})=0.918$$

$$\mathrm{Ent}(D^3)=-(\tfrac{1}{5}\log_2\tfrac{1}{5}+\tfrac{4}{5}\log_2\tfrac{4}{5})=0.722$$

- 属性“色泽”的信息增益为

$$\begin{aligned}\mathrm{Gain}(D,\text{色泽})&=\mathrm{Ent}(D)-\sum_{v=1}^{3}\frac{|D^v|}{|D|}\mathrm{Ent}(D^v)\\&=0.998-(\tfrac{6}{17}\times1.000+\tfrac{6}{17}\times0.918+\tfrac{5}{17}\times0.722)\\&=0.109\end{aligned}$$

信息增益实例

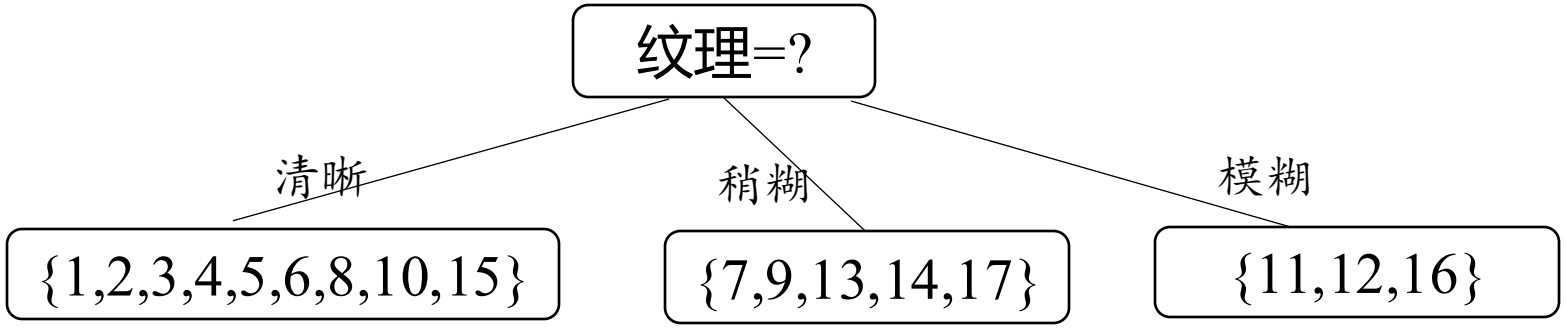
- 类似的，其他属性的信息增益为

$$\mathrm{Gain}(D, \text{根蒂}) = 0.143 \qquad \mathrm{Gain}(D, \text{敲声}) = 0.141$$

$$\mathrm{Gain}(D, \text{纹理}) = 0.381 \qquad \mathrm{Gain}(D, \text{脐部}) = 0.289$$

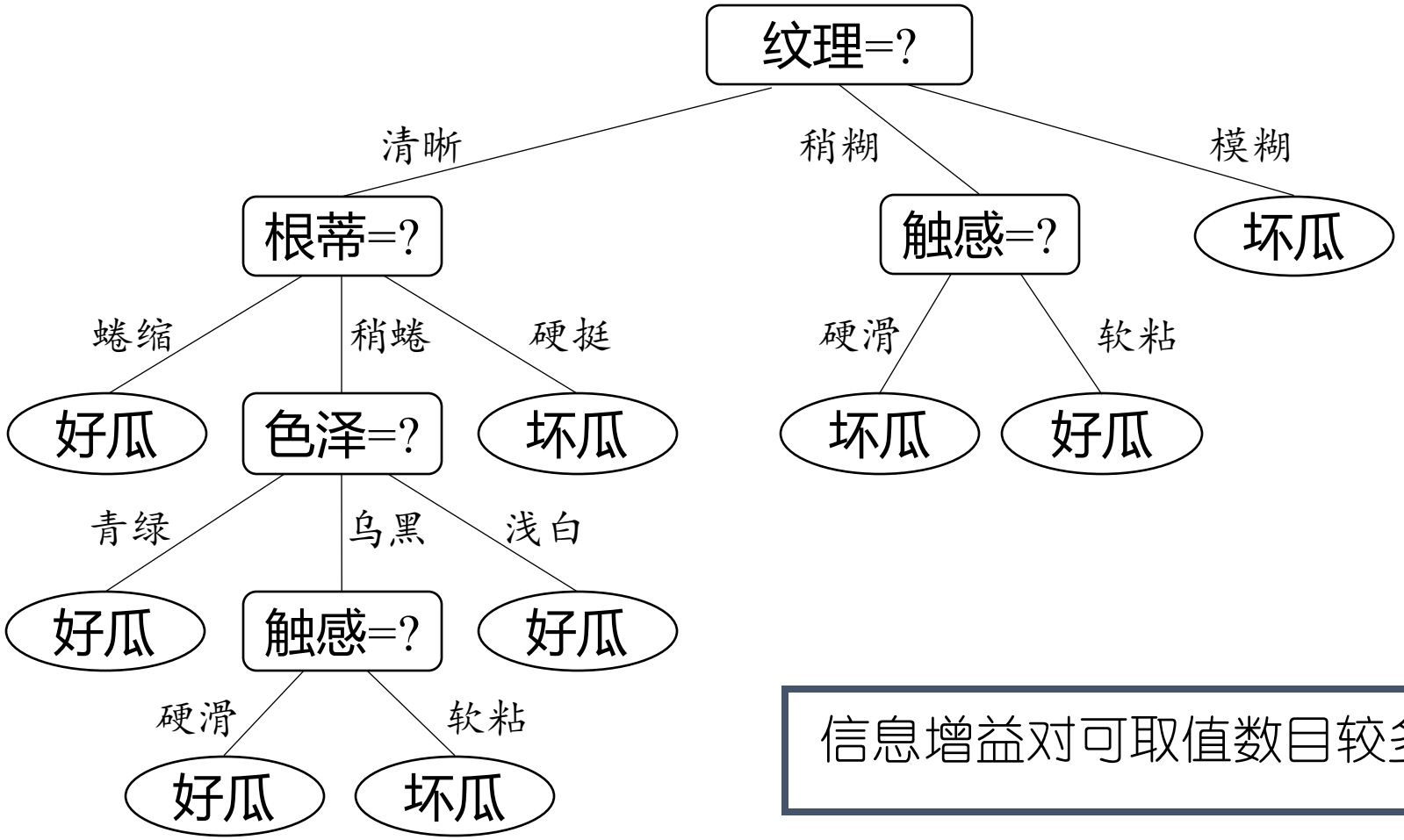
$$\mathrm{Gain}(D, \text{触感}) = 0.006$$

- 显然，属性“纹理”的信息增益最大，其被选为划分属性



信息增益实例

- 决策树学习算法将对每个分支结点做进一步划分，最终得到的决策树如图：



信息增益对可取值数目较多的属性有所偏好

划分选择-增益率

- 增益率定义：

$$\text{Gain_ratio}(D,a)=\frac{\text{Gain}(D,a)}{\text{IV}(a)}$$

其中

$$\text{IV}(a)=-\sum_{v=1}^{V}\frac{|D^v|}{|D|}\log_2\frac{|D^v|}{|D|}$$

称为属性 a 的“固有值” [Quinlan, 1993]，属性a的可能取值数目越多（即 V 越大），则 $\text{IV}(a)$ 的值通常就越大

- 存在的问题

增益率准则对可取值数目较少的属性有所偏好

- C4.5 [Quinlan, 1993]使用了一个启发式：先从候选划分属性中找出信息增益高于平均水平的属性，再从中选取增益率最高的

划分选择-基尼指数

- 数据集 D 的纯度可用“基尼值”来度量

$$\mathrm{Gini}(D) = \sum_{k=1}^{|\mathcal{Y}|} \sum_{k' \neq k} p_k p_{k'} \quad = 1 - \sum_{k=1}^{|\mathcal{Y}|} p_k^2$$

反映了从 D 中随机抽取两个样本，其类别标记不一致的概率

$\mathrm{Gini}(D)$ 越小，数据集 D 的纯度越高

- 属性 a 的基尼指数定义为：

$$\mathrm{Gini_index}(D, a) = \sum_{v=1}^{V} \frac{|D^v|}{|D|} \mathrm{Gini}(D^v)$$

- 应选择那个使划分后基尼指数最小的属性作为最优划分属性，即

$$a_* = \underset{a \in A}{\mathrm{argmin}} \ \mathrm{Gini_index}(D, a)$$

- CART [Breiman et al., 1984]采用“基尼指数”来选择划分属性

剪枝处理

- 为什么剪枝
  - “剪枝”是决策树学习算法对付“过拟合”的主要手段
  - 可通过“剪枝”来一定程度避免因决策分支过多，以致于把训练集自身的一些特点当做所有数据都具有的一般性质而导致的过拟合

- 剪枝的基本策略
  - 预剪枝
  - 后剪枝

- 判断决策树泛化性能是否提升的方法
  - 留出法：预留一部分数据用作“验证集”以进行性能评估

剪枝处理

- 数据集

训练集

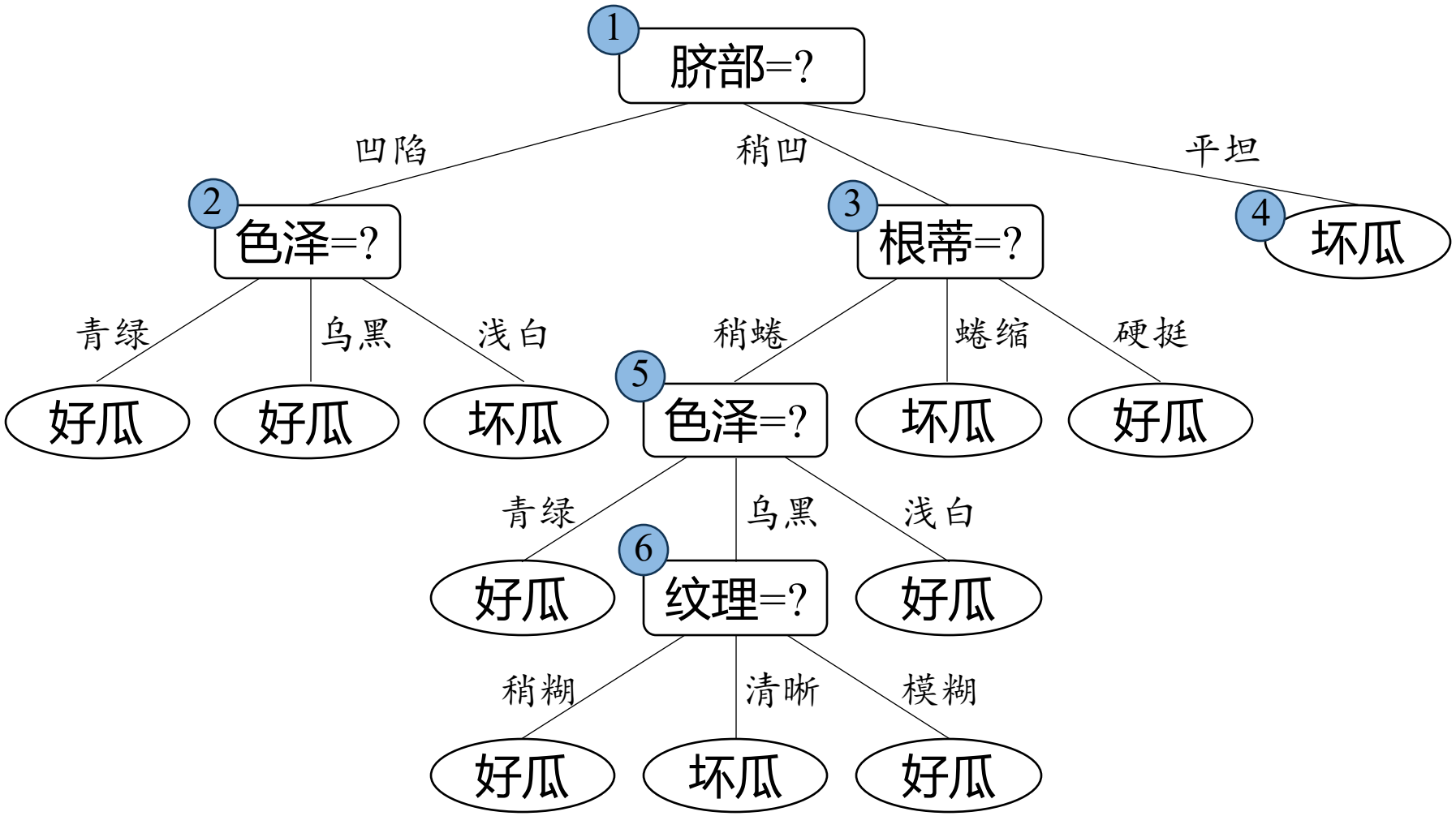
编号	色泽	根蒂	敲声	纹理	脐部	触感	好瓜
1	青绿	蜷缩	浊响	清晰	凹陷	硬滑	是
2	乌黑	蜷缩	沉闷	清晰	凹陷	硬滑	是
3	乌黑	蜷缩	浊响	清晰	凹陷	硬滑	是
6	青绿	稍蜷	浊响	清晰	稍凹	软粘	是
7	乌黑	稍蜷	浊响	稍糊	稍凹	软粘	是
10	青绿	硬挺	清脆	清晰	平坦	软粘	否
14	浅白	稍蜷	沉闷	稍糊	凹陷	硬滑	否
15	乌黑	稍蜷	浊响	清晰	稍凹	软粘	否
16	浅白	蜷缩	浊响	模糊	平坦	硬滑	否
17	青绿	蜷缩	沉闷	稍糊	稍凹	硬滑	否

验证集

编号	色泽	根蒂	敲声	纹理	脐部	触感	好瓜
4	青绿	蜷缩	沉闷	清晰	凹陷	硬滑	是
5	浅白	蜷缩	浊响	清晰	凹陷	硬滑	是
8	乌黑	稍蜷	浊响	清晰	稍凹	硬滑	是
9	乌黑	稍蜷	沉闷	稍糊	稍凹	硬滑	否
11	浅白	硬挺	清脆	模糊	平坦	硬滑	否
12	浅白	蜷缩	浊响	模糊	平坦	软粘	否
13	青绿	稍蜷	浊响	稍糊	凹陷	硬滑	否

剪枝处理

- 未剪枝决策树


1
脐部=?
凹陷
稍凹
平坦
2
色泽=?
3
根蒂=?
4
坏瓜
青绿
乌黑
浅白
稍蜷
蜷缩
硬挺
好瓜
好瓜
坏瓜
5
色泽=?
坏瓜
好瓜
青绿
乌黑
浅白
好瓜
6
纹理=?
好瓜
稍糊
清晰
模糊
好瓜
坏瓜
好瓜

剪枝处理

- 决策树生成过程中，对每个结点在划分前先进行估计，若当前结点的划分不能带来决策树泛化性能提升，则停止划分并将当前结点记为叶结点，其类别标记为训练样例数最多的类别

- 针对上述数据集，基于信息增益准则，选取属性“脐部”划分训练集。分别计算划分前（即直接将该结点作为叶结点）及划分后的验证集精度，判断是否需要划分。若划分后能提高验证集精度，则划分，对划分后的属性，执行同样判断；否则，不划分

剪枝处理实例

验证集

编号	色泽	根蒂	敲声	纹理	脐部	触感	好瓜
4	青绿	蜷缩	沉闷	清晰	凹陷	硬滑	是
5	浅白	蜷缩	浊响	清晰	凹陷	硬滑	是
8	乌黑	稍蜷	浊响	清晰	稍凹	硬滑	是
9	乌黑	稍蜷	沉闷	稍糊	稍凹	硬滑	否
11	浅白	硬挺	清脆	模糊	平坦	硬滑	否
12	浅白	蜷缩	浊响	模糊	平坦	软粘	否
13	青绿	稍蜷	浊响	稍糊	凹陷	硬滑	否

结点1：若不划分，则将其标记为叶结点，类别标记为训练样例中最多的类别，即好瓜。验证集中，$\{4,5,8\}$ 被分类正确，得到验证集精度为 $\frac{3}{7}\times 100\% = 42.9\%$

1 脐部=?

验证集精度

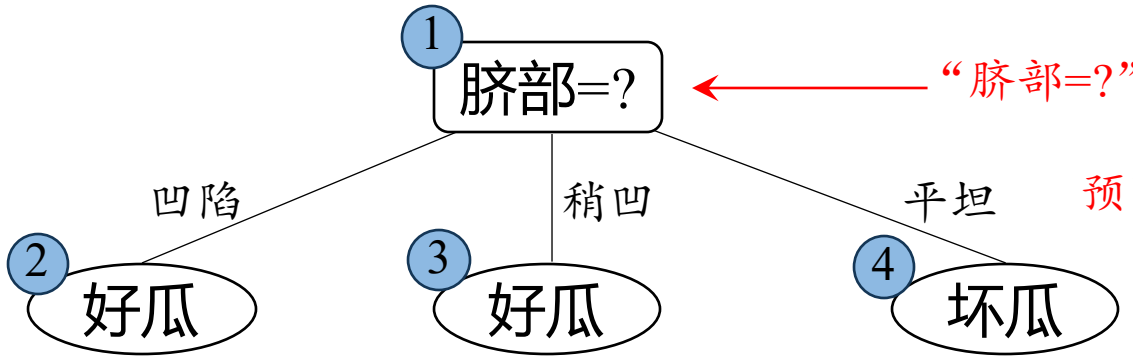
“脐部=?” 划分前: 42.9%

剪枝处理实例

验证集

编号	色泽	根蒂	敲声	纹理	脐部	触感	好瓜
4	青绿	蜷缩	沉闷	清晰	凹陷	硬滑	是
5	浅白	蜷缩	浊响	清晰	凹陷	硬滑	是
8	乌黑	稍蜷	浊响	清晰	稍凹	硬滑	是
9	乌黑	稍蜷	沉闷	稍糊	稍凹	硬滑	否
11	浅白	硬挺	清脆	模糊	平坦	硬滑	否
12	浅白	蜷缩	浊响	模糊	平坦	软粘	否
13	青绿	稍蜷	浊响	稍糊	凹陷	硬滑	否

结点1：若划分，根据结点②，③，④的训练样例，将这 3 个结点分别标记为“好瓜”、“好瓜”、“坏瓜”。此时，验证集中编号为 $\{4, 5, 8, 11, 12\}$ 的样例被划分正确，验证集精度为 $\frac{5}{7} \times 100\% = 71.4\%$



验证集精度

“脐部=?” 划分前: 42.9%

划分后: 71.4%

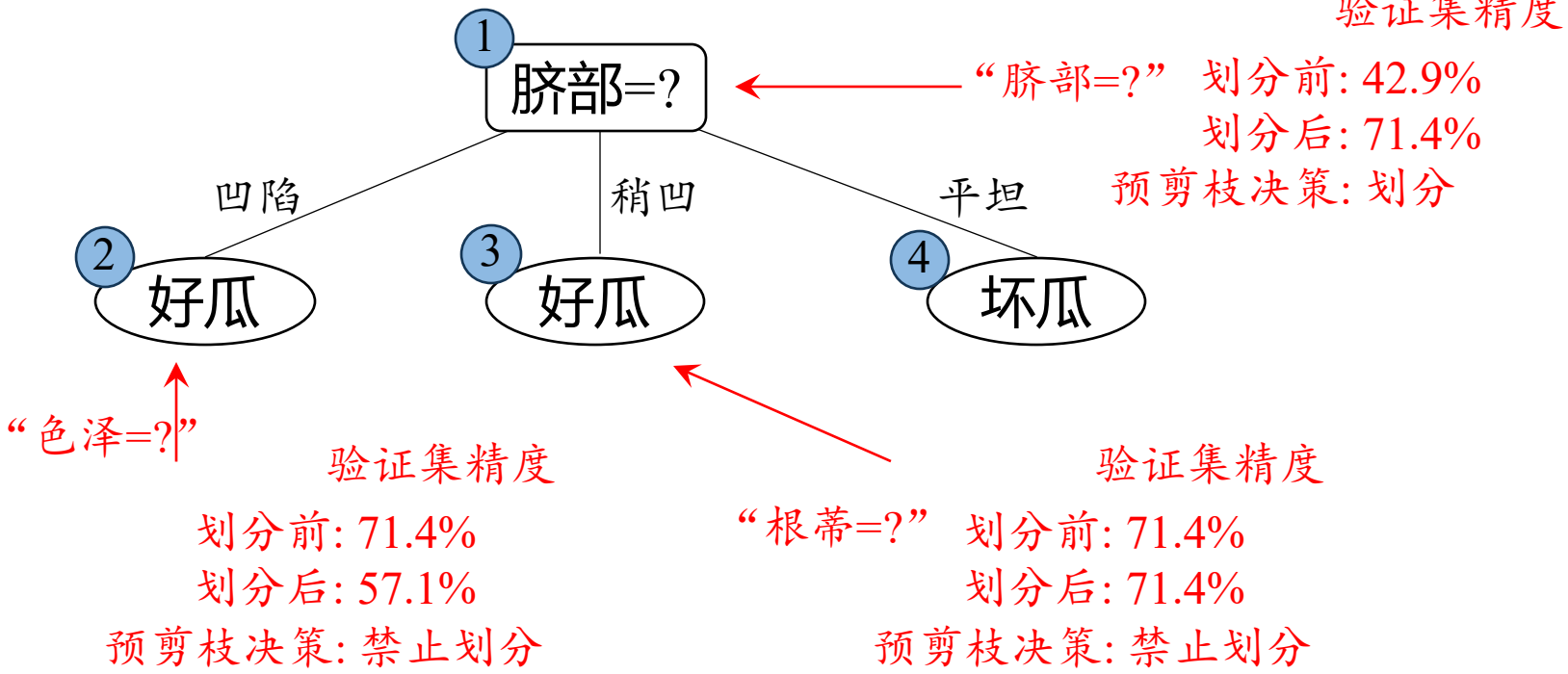
预剪枝决策: 划分

剪枝处理实例

验证集

编号	色泽	根蒂	敲声	纹理	脐部	触感	好瓜
4	青绿	蜷缩	沉闷	清晰	凹陷	硬滑	是
5	浅白	蜷缩	浊响	清晰	凹陷	硬滑	是
8	乌黑	稍蜷	浊响	清晰	稍凹	硬滑	是
9	乌黑	稍蜷	沉闷	稍糊	稍凹	硬滑	否
11	浅白	硬挺	清脆	模糊	平坦	硬滑	否
12	浅白	蜷缩	浊响	模糊	平坦	软粘	否
13	青绿	稍蜷	浊响	稍糊	凹陷	硬滑	否

对结点②，③，④分别进行剪枝判断，结点②，③都禁止划分，结点④本身为叶子结点。最终得到仅有一层划分的决策树，称为“**决策树桩**”



预剪枝的优缺点

- 优点
  - 降低过拟合风险
  - 显著减少训练时间和测试时间开销

- 缺点
  - 欠拟合风险：有些分支的当前划分虽然不能提升泛化性能，但在其基础上进行的后续划分却有可能导致性能显著提高。预剪枝基于“贪心”本质禁止这些分支展开，带来了欠拟合风险

谢　　谢

Convex optimization problem

standard form

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, i = 1, \ldots, m \\ & a_i^T x = b_i, i = 1, \ldots, p \end{array}$$

or written as

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, i = 1, \ldots, m \\ & Ax = b \end{array}$$

- $f_0, f_1, \ldots, f_m$ are convex, and equality constraints are affine.
- Feasible set of a convex optimization problem is convex, since it is the intersection of the domain of the problem, $\mathcal{D} = \cap_{i=0}^{m} \mathbf{dom}(f)$, which is a convex set, with m (convex) sublevel sets $\{x | f_i(x) \leq 0\}$ and p hyperplanes $\{x | a_i^T x = b_i\}$.

The Lagrangian

Standard form optimization problem (not necessarily convex)

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p \end{array} \tag{3}$$

with variable $x \in \mathbf{R}^n$, domain $\mathcal{D}$, optimal value p^*.

The Lagrangian

We define the Lagrangian $L : \mathbf{R}^n \times \mathbf{R}^m \times \mathbf{R}^p \to \mathbf{R}$ associated with the problem by Eq.(3) as

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{i=1}^{p} \nu_i h_i(x). \tag{4}$$

with $\mathbf{dom}(L) = \mathcal{D} \times \mathbf{R}^m \times \mathbf{R}^p$.

- λ_i: the Lagrange multiplier associated with the i-th inequality constraint $f_i(x) \leq 0$;
- ν_i the Lagrange multiplier associated with the i-th equality constraint $h_i(x) = 0$;

The Lagrange dual function

Definition

We define the Lagrange dual function (or just dual function) $g : \mathbf{R}^m \times \mathbf{R}^p \to \mathbf{R}$ as the minimum value of the Lagrangian over x: $\forall \lambda \in \mathbf{R}^m, \nu \in \mathbf{R}^p$,

$$g(\lambda, \nu) = \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) = \inf_{x \in \mathcal{D}} \left(f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{i=1}^{p} \nu_i h_i(x) \right) \tag{5}$$

Lower bound property

The dual function yields lower bounds on the optimal value p^* of the problem by Eq.(3), i.e., $\forall \lambda \succeq 0$ and $\forall \nu$, we have

$$g(\lambda, \nu) \leq p^*.$$

Least-squares solution of linear equations

The problem

$$\begin{array}{ll} \text{minimize} & x^T x \\ \text{subject to} & Ax = b, \quad A \in \mathbf{R}^{p \times n} \end{array}$$

- The Lagrangian is $L(x, \nu) = x^T x + \nu^T (Ax - b)$, with domain $\mathbf{R}^n \times \mathbf{R}^p$.
- Since $L(x, \nu)$ is a convex quadratic function of x, we can find the minimum from the optimality condition by setting gradient equal to zero:

$$\nabla_x L(x, \nu) = 2x + A^T \nu = 0 \quad \Longrightarrow x = -\frac{1}{2} A^T \nu.$$

- Plug in in $L(x, \nu)$ to obtain the dual function $g(\nu)$:

$$g(\nu) = L(-\frac{1}{2} A^T \nu, \nu) = -\frac{1}{4} \nu^T A A^T \nu - b^T \nu$$

  which is a concave quadratic function, with domain $\mathbf{R}^p$.
- lower bound property: $p^* \geq -\frac{1}{4} \nu^T A A^T \nu - b^T \nu, \ \forall \nu$.

The Lagrange dual problem

Dual problem

For each pair (λ, ν) with $\lambda \succeq 0$, the Lagrange dual function gives us a lower bound on the optimal value p^*. What is the best lower bound that can be obtained from the Lagrange dual function? This leads to:

$$\begin{array}{ll} \text{maximize} & g(\lambda, \nu) \\ \text{subject to} & \lambda \succeq 0. \end{array} \tag{8}$$

- The original problem by Eq.(3) is sometimes called the primal problem.
- The term dual feasible means, as the name implies, that (λ, ν) is feasible for the dual problem in Eq.(8).
- We refer to (λ^*, ν^*) as dual optimal or optimal Lagrange multipliers if they are optimal for the problem in Eq.(8).
- The dual problem optimum value is denoted as d^*.

Weak duality and strong duality

Weak duality: $d^* \leq p^*$

- always holds (for convex and nonconvex problems)
- can be used to find nontrivial lower bounds for difficult problems
  For example, solving the following problem (SDP):

$$\begin{array}{ll} \text{maximize} & -\mathbf{1}^T\nu \\ \text{subject to} & W + \mathbf{diag}(\nu) \succeq 0. \end{array}$$

  gives a lower bound for the two-way partitioning problem.

Strong duality: $d^* = p^*$

- does not hold in general
- (usually) holds for convex problems
- conditions that guarantee strong duality in convex problems are called constraint qualifications

KKT conditions for nonconvex problems

Assume $f_0, \ldots, f_m$, $h_1, \ldots, h_p$ are differentiable (no assumptions on convexity):

Karush-Kuhn-Tucker (KKT) conditions

Let x^* and (λ^*, ν^*) be any primal and dual optimal points with zero duality gap. Since x^* minimizes $L(x, \lambda^*, \nu^*)$ over x, it follows that its gradient must vanish at x^*, i.e.,

$$\nabla f_0(x^*) + \sum_{i=1}^{m} \lambda_i^* \nabla f_i(x^*) + \sum_{i=1}^{p} \nu_i^* \nabla h_i(x^*) = 0.$$

Thus we have the following Karush-Kuhn-Tucker (KKT) conditions:

$$\begin{aligned} f_i(x^*) &\leq 0, \quad i = 1, \ldots, m \\ h_i(x^*) &= 0, \quad i = 1, \ldots, p \\ \lambda_i^* &\geq 0, \quad i = 1, \ldots, m \\ \lambda_i^* f_i(x^*) &= 0, \quad i = 1, \ldots, m \\ \nabla f_0(x^*) + \sum_{i=1}^{m} \lambda_i^* \nabla f_i(x^*) + \sum_{i=1}^{p} \nu_i^* \nabla h_i(x^*) &= 0, \end{aligned} \tag{9}$$

For any optimization problem with differentiable objective and constraint functions for which strong duality obtains, any pair of primal and dual optimal points must satisfy the KKT conditions.

KKT conditions for convex problems

KKT conditions for convex problems

When the primal problem is convex, the KKT conditions are also sufficient for the points to be primal and dual optimal. In other words, if f_i are convex and h_i are affine, and $\widetilde{x}, \widetilde{\lambda}, \widetilde{\nu}$ are any points that satisfy the KKT conditions:

$$\begin{aligned} f_i(\widetilde{x}) &\leq 0, \quad i = 1, \ldots, m \\ h_i(\widetilde{x}) &= 0, \quad i = 1, \ldots, p \\ \widetilde{\lambda}_i &\geq 0, \quad i = 1, \ldots, m \\ \widetilde{\lambda}_i f_i(\widetilde{x}) &= 0, \quad i = 1, \ldots, m \end{aligned}$$

$$\nabla f_0(\widetilde{x}) + \sum_{i=1}^{m} \widetilde{\lambda}_i \nabla f_i(\widetilde{x}) + \sum_{i=1}^{p} \widetilde{\nu}_i \nabla h_i(\widetilde{x}) = 0,$$

then $\widetilde{x}$ and $(\widetilde{\lambda}, \widetilde{\nu})$ are primal and dual optimal, with zero duality gap.